Towards Interactive Gesture Control of Sound Patterns Using the Wiimote™

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Abstract

This paper discusses research towards an interactive gesture based control strategy for sound event patterns or sequences. It posits a direction, which seeks to nuance such sound structures, more commonly thought of as fixed, with traits reminiscent of "live" performance. Such rigid formalism is common in many music sequences programs and while a great deal of effective music has been produced in this manner to date, the option of a range of control over sequenced sound introduces the possibility of an entirely new performance and aesthetic level.

Consider repetition in contemporary electronic music, which has become an essential and extensively used structural device. In Jazz, the practice of repetition is particularly sophisticated because it is directly under the control of the musicians who understand how to exercise variation and who can carefully modulate between direct statement and complex variation. One need only think of certain groove patterns in the hands of skilled players.

In seeking a similar relationship with computer-generated sound sequences, the research described here uses established real-time software, SuperCollider 3 (SC3) and a Nintendo Wiimote™ controller from the Games world as the physical interface. In using the Wiimote, it is anticipated that the performer might have a more physical relation to the production and deployment of sound in sequences. The objective is to establish a dynamic/ theatrical relationship with complex sound structures through sensor data derived from human movement.

Introduction

Personal precedent for this research is founded in the work with *HyperSense Complex*[1] (Riddell 2005) around 2003–4. In the process of creating new performance approaches for this group, Simon Burton explored a strategy for working with sequences in an interactive manner. Although idiosyncratic, it constituted a response to developing performance approaches for this new interactive context.

Burton's idea was based upon cycling temporal placeholders or "buckets", like defined spaces on a sequential conveyor belt (initially silent buckets)[2]. These buckets could have sound event placed in them by the performer. (see Figure 1.)

When the performer triggered an event, it was placed in the nearest temporal slot in the cycle which up to that point the performer may not have either heard or appreciated was there. If the event coincided with slot one, the start of the cycle, it was given greater emphasis (attack characteristics and amplitude). This was most noticeable when the sound had naturally acute attack characteristics.

In simply bending a finger, the performer could place in those slots sound samples, which could then be manipulated in a variety of ways, such as by pitch or amplitude change or even simply muting by similar gestures with other fingers. Amplitude variation provided a very subtle but effective way to nuance a sequence. The performer could also remove events in the same way by bending a finger.

Further to this temporal activation of sound, Burton added the idea of branching, which was the subdivision of time below the initial cycle durations. So it became possible to create more complex rhythmic patterns for one performer.

```
Level 1 - 1     2     3     4     5    1,
Level 2 – 1  2  3  4  5  6  7  8  9 10 1,  etc.
```

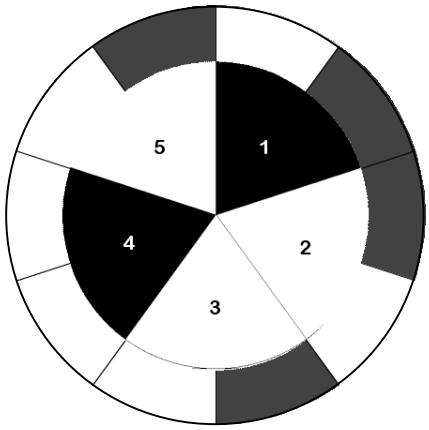


Figure 1. Diagram of Burton's "Silent Bucket" concept

The performer could thus add or subtract events from the sequences at two temporal levels. In retrospect, this composition process was challenging to engage. It required concentration on one's own patterns while considering the sum of the parts aesthetically. Activating and removing

[1] The performance trio *HyperSense Complex* used Flex sensors on the fingers to control sounds and their behaviour. *HyperSense* was a networked ensemble sharing the same audio processing space (see arrowtheory.com/hypersense/index.html).

[2] The work became known as "Drumming Tree".

events was frequently arbitrary. Precise management of events took considerable concentration in a context that could be externally influenced through effects imposed by one of the other performers.

Burton's approach was the result of thinking through a particular performance concept. Since there was no other option other than to use the sensors at our disposal and which in themselves, formed the central platform of our performance agenda, it was just assumed that the concept would engage that type of sensor and associated performance skill.

The idea of silent placeholders that could be filled with sound events was novel. It did provide some cognitive reassurance that a stable and predictable world existed into which one we could develop an improvisatory position. Even if we could not initially hear the temporary framework, it quickly became evident in the early stages of a performance.

If this approach had been for only one performer, the idea would have perhaps needed expansion or augmentation to include a greater range of parameters to expand the greater diversity in the compositional domain. This would likely have happened with time, practice and musical evolution, as is the case of all musical instrument development.

However, the initial implementation was realised through three performers acting simultaneously and in this respect, the relative simplicity of the system was obscured by a complexity of three performers aspiring to a unified sound. Three performers, each with differing sounds and effects, could create a compelling ensemble effect and a more engaging sound experience.

The network performance configuration, discussed elsewhere (Riddell 2005) and explore by artists like Phil Burke (Burke 2000) in recent years, remains an intriguing approach to the contemporary idea of the ensemble with many variations possible. While not explicitly discussed in this paper, it remains a logical extension to the practice described here.

So the previously described personal experience tangibly underpins this research into gesture and musical performance but not to the exclusion of other research the findings of which are presented at many conferences (CHI and NIME, etc). What is interesting is that much of the research seeks to explore the area through similarly idiosyncratic means, which it could be argued parallel the development of musical instruments in general. The body is a primary and logical source of expression as discussed in Strachan et al (Strachan 2007), who outline an approach to controlling a music player. Hashimoto et al (Hashimoto 2005), take a more conventional position, which considers that "in traditional musical instruments, the relationship between the action and the generated sound is determined by the physical structures of the instruments." There is some logic in that statement but Goto and Suzuki (Goto 2004) engage virtual musical instruments and sound synthesis in an attempt to explore gesture and performance beyond traditional instruments. These researchers, in part, circumscribed directions in gesture and musical performance currently in vogue.

Research Agenda

In practical terms, the research direction taken here differs significantly from that of the *Hyper-Sense Complex* project in both technology and anticipated creative outcome. The principle interest here is the extent to which patterns or sequences of sounds can be manipulated with arm based gesture control of a different physical character.

The research here does hold, as a fundamental position, the ideal of human instrumental performance but additionally recognizes that the mapping of such behaviour as that which lends traditional music a great sense of humanity, may not be possible or even desirable in respect to complex computer generated events. This research does, therefore, acknowledge an ambition to explore uncharted areas of interaction in the digital sound domain.

At the core of the approach is the use of sophisticated pattern generating systems and subsequent functionality associated with the inherent messaging system of the audio application, *SuperCollider 3*. It is recognized that other applications of a similar status (MAX, PD, Csound, etc.) could possibly be used as instead.

Technology

This research exploits the sophistication of Nintendo's Wiimote controller, which is extensively documented on the Internet (sites in reference section) and contains the following functionality:

Sensors
- Depth
- Tilt
- Position

Buttons
- 1 Trigger
- 6 Buttons
- 4 way tilt

Features
- Infrared receiver for depth and position
- Speaker for depth-of-sound
- Rumble motor
- 4 Led Display

Table 2. WiiMote Functions

The commercially available Wiimote functions using *BlueTooth*™ communication technology and possesses a unique physical design that has been researched by Nintendo. Admittedly, the intended use of the Wiimote is for computer games that require a more physical interaction but exactly what is possible at a subtle level of creative sound engagement is yet to be fully appreciated.

From the above list of functions and internal operations, it can be appreciated that the Wiimote

has considerable potential as a sound performance device. This can be understood through the proliferation of such applications for the Mac, PC and Linux systems as *DarwiinRemote*, *Glovepie*, *Remote Buddy*, *WiiViewer* and *WiinRemote*.

This research is primarily interested in the Wiimote implementation for *SuperCollider* (SC3) developed by Pete Moss. However, it is worth noting that there exists a *Wiimote-api* which is "a library written in C to access the Wii Remote's abilities" (software reference). This offers a near complete set of functions pertaining to the Wiimote for those who wish to develop a different strategy for using the Wiimote.

Wiimote and SC3

The following discussion is based on Pete Moss' Wiimote implementation for SC3, which consists of the classes: `Wiimote.sc, WiimoteUgens.sc` and the plugin, `Wiimote.scx`. There is basic functionality for all the buttons and X, Y and Z movement, Rumble and LEDs. Help is also included in the distribution. Wiimote functionality is represented in the following Ugens:

```
WiimoteX        accelerometer Ugen
WiimoteY        accelerometer Ugen
WiimoteZ        accelerometer Ugen
WiiMotion       Motion Ugen
WiiTrig         Motion Trigger Ugen
Wiimote1        Button "1" Ugen
Wiimote2        Button "2" Ugen
WiimoteA        Button "A" Ugen
WimoteB         Button "B" Ugen
WiimoteMinus    Button "-" Ugen
WiimotePlus     Button "+" Ugen
WiimoteHome     Button home Ugen
WiimoteUp       Tilt up Ugen
WiimoteDown     Tilt down Ugen
WiimoteLeft     Tile Left Ugen
WiimoteRight    Tilt Right Ugen
WiiRumble       Rumble motor Ugen
WiiLED          LED Ugen
```

Table 3. Ugens available in Wiimote.sc

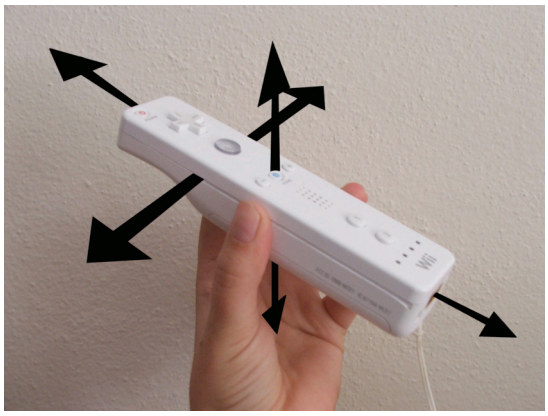


Figure 2. WiiMote Axes

So the Wiimote Ugens are controllers defined in the form:

```
WiimoteX.kr()
WiimoteY.kr()
WiimoteZ.kr()
```

Ex. 1. X, Y and Z movement Ugens

Here control data received from the X and Y axis in the range 0 –1.0.

This nomenclature defines the basic operation of the key functions of the Wiimote. It is curious to reflect on the idea of using vibration as a means of experiencing a kind of haptic response to event triggering. This functionality is not represented in the following examples but would probably be easy enough to implement to, for example, delineate usable ranges of the X, Y and Z parameters.

It can be appreciated from the above description that the SC3 Ugens are control rate and so are used only inside SynthDefs, that is, on the server side of SC3. This functionality will not be discussed in detail here because it is not pertinent and also more or less self-evident from the examples for the Ugens.

Connectivity

After following the Wiimote initialization procedure control data is available in the SynthDefs. Wiimote data at this point is not useful for the pattern structures and will need to be processed through an OSCresponder, which facilitates client-side data in the following manner:

```
SynthDef(\wiiX, {
   SendTrig.kr(Impulse.kr(10),
   13,LinLin.kr(WiimoteX.kr(),
   -0.17,0.2,0.4,2.0));
}).load(s);
)
s.sendMsg("/s_new", "wiiX", node =
s.nextNodeID,1,1);

OSCresponder(s.addr,'/tr', {| t,r,msg|
// SendTrig data can be accessed here.

}).add;
```

Ex. 2. Client side "X" axis data

Now that we have data on the client side, it is possible to pass it to a pattern structure under certain conditions. In this particular instance, SendTrig is passing the control data from the "X" axis of the Wiimote to the OSCresponder at the rate of 10 messages per second. What the rate should be depends on the use of the data in the pattern structure. It would be expected that the pattern would be accessed in an OSCresponder as:

```
OSCresponder(s.addr,'/tr', { arg
time,responder,msg;
if ((msg[1] == node), {
   amp = msg[2];
   h = [0.125,Prand([0.1, 0.5, 1.0],
   2),0.5,0.125];
   j = Pseq(h, inf);
   k = Pseq((0..7) + 15.rand;
   Pdef(\x, Pbind(\dur, j, \amp, amp, \degree,
k, inf)));
});
}).add;
```

Ex. 3. A pattern accessed on the client side

Defining Pattern Control

Given the two control data access modes, continuous data or toggle state data, pattern structures should be able to use data from at least one. This leads to the question of what is the form of a pattern structure? Consider the following, which represents a basic structure defined through the class Pbind. This assumes an instrument called "someInstrument" which takes arguments defined by the names preceded by a "\".

```
r = Pbind(
   \instrument, 'someInstrument',
   \dur, Pseq([0.5,0.8,0.8,0.4,0.2],inf),
   \amp, 0,
//Prand([2.0,6.0,5.0,7.0,7.0,6.0,5.0,7.0],inf),
   \rate, Prand([0.6,Pseq([-1.5,0.7,-
   0.6,0.8,0.6,-1.5],Prand([3,4.0],1))],inf),
   \startpos, 109,
   \bufnum, 1,
   \panrate, Prand([0.9, 1.0, 0.5,0.3, 0.1,
   3.0, 2.0],inf),\release, 0.95
);
g = r.play; // This plays the above structure
```

Ex. 4. A pattern structure

The point to note here is that while this structure defines an initial state of the pattern it can be modified by making it a Stream and re-initializing it. This example shows the next stage in the creation of an interactive pattern:

```
//  g
g.stream = Pbind(
   \instrument, 'someInstrument',
   \amp,
   Prand([5.0,3.0,7.0,4.0,7.0,1.0,5.0,7.0],inf)
   ,
   \dur,
   Pseq([0.5,0.2,0.3,0.4,0.2,Pseq([0.15],6)],in
   f),
   \startpos, 107,
   \release, 0.95,
   \bufnum, 1,
   \panrate, Prand([0.9, 1.0, 0.5,0.3, 0.1,
   3.0, 2.0],inf),
   \rate, Prand([0.6, Pseq([-1.5,0.7,-0.6,-
   0.8,0.6,-1.5],Prand([3,4],1))],inf)
).asStream;
```

Ex. 5. A pattern structure as a Stream

Data for any of the second elements of the duple , e.g. `\release, 0.95` can be modified and the structure selected and executed in the conventional SC3 manner. Thus dynamically changing the structure.

At this point, however, there is no connectivity with the Wiimote nor is there provision for changing the content of the pattern structure "g".

Reflecting on the initial objective of this research, the primary purpose was to modify pattern repetition in such a way that retained certain characteristics while introducing variation of a particular type.

Basic functionality includes the ability to start, mute and terminate sequences as well as vary internal parameters.

The use of the Wiimote invariably leads to considerable data input because the fundamental operation is as a continuous controller. This is necessary for anything that is expected to operate in real-time. The question here is can such a continuous controller provide data to nuance patterns and still feel to the user like a real-time interactive device? This question will become more pertinent at the end of this paper.

The operational characteristics of the Wiimote suggest that discrete control data requires skilled use. Finely grained values are the outcome of careful control behaviour.

```
SynthDef(\wiiX, {
   SendTrig.kr(Impulse.kr(10),
   0,LinLin.kr(WiimoteX.kr(),-0.175,
   0.23,0.0,0.9));
}).load(s);

// Below here just to show initialization
s.sendMsg("/s_new", "wiiY", node =
s.nextNodeID,1,1);

OSCresponder(s.addr,'/tr', {|
time,responder,msg|
if ((msg[1] == node), {
   amp = msg[2];
   h = [0.125,Prand([0.1, 0.5, 1.0],
   2),0.5,0.125];
   j = Pseq(h, inf);
   k = Pseq((0..7) + 15.rand;
   Pdef(\x, Pbind(\dur, j, \amp, amp, \degree,
k, inf)));
});
}).add;
```

Ex. 6. Client side data used to control amplitude

The above example suggests that the amplitude of the pattern is being controlled not the individual sounds of the pattern. The difference, if that is the case is not significant and the more efficient way to do that is directly in the SynthDef.

Using variable controller data, not just toggle state data, can be handled as:

```
SynthDef(\wiiX, {
   SendTrig.kr(Impulse.kr(10),
   0,LinLin.kr(WiimoteX.kr(),-
   0.175,0.23,0.0,0.9)
   );
}).load(s);

// Below here just to show initialization
s.sendMsg("/s_new", "wiiY", node2 =
s.nextNodeID,1,1);

h = Array.fill(rand(13) + 1,{rrand (0.0125,
2.0).round(0.01)});
asize = h.size;
k = Pseq(((0..asize) * 50) + 300, inf).postln;

patfunc = {|parg, asize|
   if (parg > 0,{h[rand(asize)] = parg;});
   j = Pseq(h, inf);
};

patfunc.value(1,asize);
Pdef(\x, Pbind(\instrument, \theSynth, \dur, j,
\amp, 1.0, \ffreq, k));

OSCresponder(s.addr,'/tr',
{|time,responder,msg|
```

```
// Y axis
if ((msg[1] == node2), {
   if(running == 1, {
      patfunc.value(msg[3], asize);
      n = k.next;
      Pdef(\x).set(\dur, j);
      Pdef(\x).set(\ffreq, n);
   });
});
}).add;
```

Ex. 7. "Y" values inserted into a sequence

In the above code, the "Y" axis values are randomly inserted into the duration sequence. The length of the pattern is randomly determined at the start. The pattern gets updated 10/s but that could be slower and made variable by updating the SynthDef to change the Impulse trigger rate.

Buttons can be used for selective operations:

```
SynthDef(\wiiB, {
   SendTrig.kr(WiimoteB.kr(0.0),0,0.0);
}).load(s);

if ((msg[1] == node8), { // Button B
   h = Array.fill(rand(13) + 1,{rrand (0.0125,
   2.0).round(0.01)});
   asize = h.size;
   j = Pseq(h, inf);
   k = Pseq(((0..asize) * 50) + 300, inf);
   Pdef(\x, Pbind(\instrument, \theSynth, \dur,
   j, \amp, 1.0, \ffreq, k));
});
```

Ex. 8. A pattern changed through button activation

The above code allows button "B" to create a new pattern during performance.

The discussion here has focused on the operational state of the Wiimote and has not taken into account any skill level or experience. It is understood that this will have a considerable influence on the development of these operations.

Temporal Nuance

The tempo rate and amplitude changes afforded by movement in the X and Y axes exercise some degree of influence on a running sequence. This may not be particularly exciting but that could depend on the chosen temporal structure and sounds used. However, to further add to this chemistry of action, "Z" axis data could be used to create subtle changes in the timing of the structure.

The primary question is what are subtle changes to the timing of the structure? Let's initially assume that these changes are very small and that we want to alter the sequences based on how many elements there are in the sequence. Further subtlety might be found in dynamically changing the attack points of certain critical events within the structure. This assumes a recognizable metric structure like 4/4, etc. Returning to the build structure function (Ex. 9), something to generate regular metrical events could be constructed by:

```
h = Array.fill([2,4,8,16].choose,
{1/[2,4,8,16].choose});
```

Ex. 9. Generating a metrical structure with durations

The array "h" could look like one of these:

```
[0.5, 0.5]
[0.5, 0.5]

[0.25, 0.25, 0.0625, 0.0625]
[0.25, 0.25, 0.0625, 0.0625]

[0.0625, 0.125, 0.125, 0.0625, 0.5, 0.0625,
0.25, 0.25, 0.125, 0.25, 0.5, 0.5, 0.25, 0.5,
0.0625, 0.125]

[0.0625, 0.125, 0.125, 0.0625, 0.5, 0.0625,
0.25, 0.25, 0.125, 0.25, 0.5, 0.5, 0.25, 0.5,
0.0625, 0.125]
```

Ex. 10. Arrays generated by Ex. 10

These divisions of 1 provide the basis for a temporal structure that can be subtly modified to reduce the consistency of the values.

Taking this further, the author has reviewed another technique for sequence variation based on Jaffe's (Jaffe 1985) time map method of sinusoidal tempo oscillation and more recent work by Collins (Collins 2003).

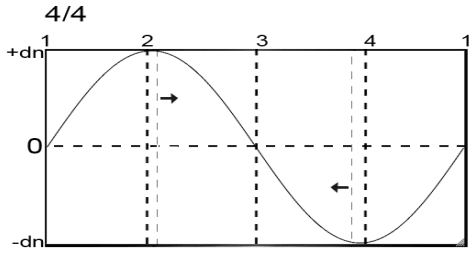


Figure 3. A simple sinusoidal mapping onto 4/4

The application of sinusoidal data to timing information within a defined temporal sequence seeks to achieve temporal displacement through dn augmentation, within that time frame but ensure correct overall temporal accuracy between sequences. In this case, a rhythmic value can be altered by the addition of a corresponding sine value.

```
var inc = 0;
d = 2pi/512; // Two PI and resolution
// create an empty array
a = Array.newClear(0);
512.do({|i|
// fill with Sine data
   a = a.add(sin(inc));
   inc = inc + d;
});
```

Ex. 11. Filling an array with Sine data

Having stored signal data in an array, it is just a question of using the data to compute certain points against the incoming data. This data is part of a process to converted deviation time, which can be added or subtracted to the event time to a performance out come. The event time is calculated in terms of the total sequence duration. Although this description is rather simplistic, it provides the foundation of a result for corrected deviations that maintains temporal integrity of the sequence duration.

Such temporal shifts in the rhythm domain seek to emulate not only the underlying stylistic and effect foundations of music such as Jazz but music of other world cultures where temporary displacement is an operational component in defining the power of the music. This technique and several others, which aspire to the same functionality, have been discussed in more technical detail in McGuiness (2004). However, they do not take into account the idea of influencing extant patterns by performance interaction.

Future Directions

The idea of a "pattern" has until this point been defined loosely as the repetition of a collection of events, possibly 3 or more, with the potential of degrees of variation that, for the most part, do not radically alter the pattern structure. The sequence is maintained and recognizable.

Now the function of the pattern will be reconsidered in terms of a sub-structure that is perhaps not audible as a singular repeatable entity, definitely exists but is not necessarily directly repeated. This concept has been understood for many years and has been manifest by various musicians, most notably in the improvisatory solos of Charlie Parker:

> "Still enriching the jazz vocabulary half a century after they came into being, the components of the musical legacy of Charlie Parker are his "personal repertory of melodic formulas that he used in the course of improvising." Owens points out that every noteworthy improviser puts together and reworks "a well-rehearsed bag of melodic tricks....His well-practiced melodic patterns are essential identifiers of his style." Earlier, Owens states that "...his 'spontaneous' performances were actually precomposed in part. This preparation was absolutely necessary, for no one can create fluent, coherent melodies in real time without [it]...."
>
> Of course, there's more than a constant re-combination of patterns and "licks" that makes Charlie Parker's solos so remarkable. Beyond these, and after acknowledging the use of musical "quotes" from sources popular to classical, factoring in his superb musicianship, his "sheer energy and self-confidence," Professor Owens himself reaches a thematic crescendo by detailing "...another factor that helps generate the sense of rightness in his music. Typically entire phrases, and even entire choruses and groupings of choruses, are goal-oriented; they arrive on a final note that lies at the end of a lengthy stepwise descent." This downward-scaling structure was something Parker alone contributed to jazz, perhaps the capstone of his great gift to the art form."(Simon).

The "precomposed" formalism is a strong theory founded in the plausible outcome of a lifetime immersed in an evolving Jazz idiom. The question pertinent to this research is, how might this apply to the Wiimote performance practice?

It will be recalled that an objective of this research was to imbue patterns with a sense of an organic form, structured but individual. The idea of repetition becomes crucial to the aesthetic architecture of the sound statement. Given Parker's talent, genius and musical context, expectations of a practice, using the Wiimote, achieving anywhere near as refined state are premature to say the least. However, the idea of a concatenation of patterns that have some kind of pre-configuration is a logical direction for a device with such a diverse control output.

So implicit in this approach is the establishment of patterns prior to performance. These could perhaps be rehearsed but the immediate question is, how might a performer access them?

In performance, the conscious selection and awareness of patterns in action, would prove to be too inhibiting in the pursuit of a coherent and aesthetically engaging continuity. The sound of the patterns should be the only guide to how they are deployed.

This suggested a rethinking of Simon Burton's idea of silent temporal spaces into which sounds are placed by the action of the performer. Suppose that all the patterns where simply place markers and that the performer's actions claimed them in a temporal space? The challenge lies in the substance of and access to, the performance data.

If the Wiimote data could be used to place sounds in patterns by approximation to the event space, then there may need to be system that selects the patterns themselves. The other problem with this is the sense of real-time engagement. This approach is contributory in nature not immediately interactive.

This direction suggests the possible use of L-Systems (McCormack 1996) in which a grammar could be use to define the construction and filling of patterns according to data received from the Wiimote. While this research has yet to be undertaken it does offer the possibility of formalizing pattern generation in the context of silent temporal units. The obvious problem is that the sense of response in the creation of new patterns or the modification of existing ones would possibly be compromised. To ameliorate this situation, the Wiimote could function on two levels. First, there would be the capacity to generate patterns and modify existing ones. Secondly, allow a more responsive interaction through direct manipulation of SynthDef Ugens. These could work in combination to produce an

effective system for interaction. Although challenging to learn.

Conclusion

The Wiimote has, in its brief existence, attracted considerable interest, examination and redeployment as a controller. It offers useful and interesting functionality in a small package—yet to be fully explored.

The research outlined here has suggested a few modes of engagement to influence sequence structures and their performance. As the structures become more complex, use of the device becomes something that inevitably requires training and practice, just as a traditional instrument does.

More importantly, the gestures become subtle or as expressed at the end of this paper, abstracted. If one were looking for a device where large-scale physical movement of the arms and body in general, resulted in a diversity of sound, then one would be disappointed. Working with patterns/sequences requires listening and responding, which is part of a skill that has to be acquired over time and creative outcomes. Grander theatrical gestures would seem to emerge from mastery of those refined subtle movements that generate real sound nuances.

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Wii + Supercollider www.youtube.com/watch?v=z2Z1Pvt8Poc (accessed 08-04-07)